類別資料分析 第3章 (2)

The Analysis of Categorical Data

Chapter 3 (2) : Generalized Linear Models

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Generalized Linear Models

*Y_i : the response variable = the random component; $\mu_i = E(Y_i)$

*X_i : the explanatory variables = the systematic component

$\alpha + \beta_1 x_1 + \cdots + \beta_k x_k$: the linear predictor

*$g(\mu) = \alpha + \beta_1 x_1 + \cdots + \beta_k x_k$, the function $g(\cdot)$ is called the link function

(1) the identity link : $g(\mu) = \mu = \alpha + \beta_1 x_1 + \cdots + \beta_k x_k$

(2) the log link : $g(\mu) = \ln \mu = \alpha + \beta_1 x_1 + \cdots + \beta_k x_k$: log linear model

(3) the logit link : $g(\mu) = \ln \frac{\mu}{1-\mu} = \alpha + \beta_1 x_1 + \cdots + \beta_k x_k$: logit model

(4) the probit link : $g(\mu) = \Phi^{-1}(\mu) = \alpha + \beta_1 x_1 + \cdots + \beta_k x_k$: probit model

Generalized Linear Models

· If $Y \sim N(\mu, \sigma^2) \Rightarrow f(y; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-\mu)^2}{2\sigma^2}} = \exp\left\{-\frac{y^2}{2\sigma^2} + \frac{y}{\sigma^2}\mu - \frac{\mu^2}{2\sigma^2} - \frac{1}{2}\ln(2\pi) - \frac{1}{2}\ln\sigma^2\right\}$; $-\infty < y < \infty$

the canonical link : $g(\mu) = \mu$: the nature parameter

· If $Y \sim Poi(\mu) \Rightarrow f(y; \mu) = \frac{e^{-\mu}\mu^y}{y!} = \exp\left\{-\mu + y\ln\mu - \frac{1}{2}\ln(y!)\right\}$; $y = 0, 1, 2, \ldots$

the canonical link : $g(\mu) = \ln(\mu)$

· If $Y \sim Bin(n, \mu) \Rightarrow f(y; \mu) = \mu^y(1-\mu)^{n-y} = \exp\left\{y\ln(\frac{\mu}{1-\mu}) - n\ln(1-\mu)\right\}$; $y = 0, 1$

the canonical link : $g(\mu) = \ln(\frac{\mu}{1-\mu})$

*分析非常態資料傳統方法：將資料轉換成近似常態，並具有相等變異數

e. g. Box-Cox transformation, stabilize variance

Generalized Linear Models

*Binary data 利用 linear probability model (ordinary regression) 分析的缺點：

(1) $\mathbf{Y \sim Ber(\pi), \pi = \pi(x) = \alpha + \beta x \Rightarrow \hat{\pi}(x) = \hat{\alpha} + \hat{\beta} x}$

$\mathbf{P(Y = 1) = \pi, P(Y = 0) = 1 - \pi}$

(2) $\mathbf{E(Y) = \pi, 0 < \pi < 1}$ （當 x 很大或很小，會使 $\mathbf{\hat{\pi}(x) < 0}$ 且 $\mathbf{\hat{\pi}(x) > 1}$ ）

(3) $\mathbf{E(Y) = \pi(1 - \pi)}$ is not constant for all x.

雖然可作 Generalized least square，但並非較好的分析方法。

*GLM 利用概似函數來求 link function中，$\boldsymbol{\alpha}$ 和 $\boldsymbol{\beta}$ 的估計量 (MLE)，

因此不須 Normal 和 Variance is constant 的假設。

又因 MLE 通常沒有具體的公式 (Closed-form)，故須藉由計算機利用數值分析方法求解。

Generalized Linear Models

e. g. 流行病學研究打鼾是否為心臟病的危險因子

x_i score	Snoring	Heart Disease		Proportion Yes	Linear fit	Logit fit	Probit fit
		Yes	No				
0	Never	24	1355	0.017	0.017	0.021	0.020
2	Occasional	35	603	0.055	0.057	0.044	0.046
4	Nearly every night	21	192	0.099	0.096	0.093	0.095
5	Every night	30	224	0.118	0.116	0.132	0.131

Generalized Linear Models

e. g. 流行病學研究打鼾是否為心臟病的危險因子

(1) Linear fit : $\pi = \pi(x) = \alpha + \beta x \Rightarrow \hat{\pi}(x) = \hat{\alpha} + \hat{\beta}x$: the fitted value

MLE : $\hat{\alpha} = 0.0172, \hat{\beta} = 0.0198, \text{s.e.}(\hat{\beta}) = 0.0028$

1. $H_0 : \beta = 0$ vs. $H_1 : \beta \neq 0$

2. $Z = \frac{\hat{\beta}}{\text{s.e.}(\hat{\beta})} = \frac{0.0198}{0.0028} \sim N(0,1)$ under H_0, $Z^2 \sim \chi_1^2$

Z: Wald statistic

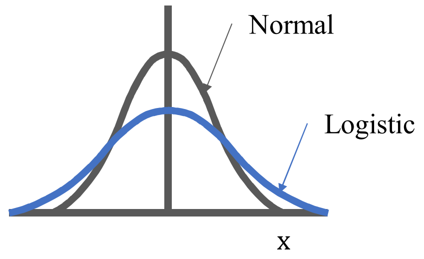
3. Reject $H_0 \because Z^2 = 49.9728 > \chi_{1,\alpha}^2 = 3.84$

· Using ordinary least squares : $\hat{\alpha} = 0.0169,\ \hat{\beta} = 0.02$

Generalized Linear Models

e. g. 流行病學研究打鼾是否為心臟病的危險因子

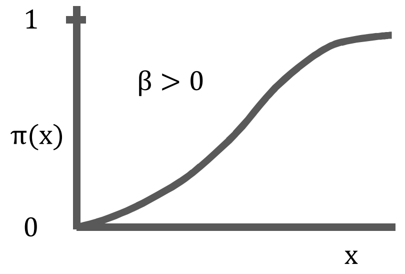
(2) Logit fit : $\ln\frac{\pi(x)}{1-\pi(x)} = \alpha + \beta x \Rightarrow \hat{\pi}(x) = \frac{1}{1+e^{-(\hat{\alpha}+\hat{\beta}x)}}$



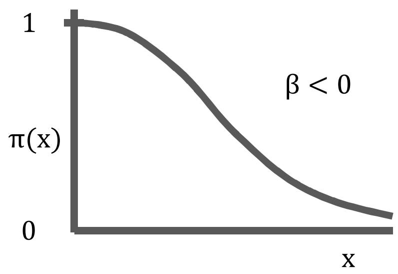
MLE : $\hat{\alpha} = -3.87, \hat{\beta} = 0.4, s.e.\left(\hat{\beta}\right) = 0.05, \text{Wald } Z^2 = 63.12$

· Let X be a random variable, if X~two parameter logistic distribution

$$\Rightarrow F(x) = P(X \le x) = \frac{1}{1+e^{-(\alpha+\beta x)}} \text{ or } \frac{e^{\alpha+\beta x}}{1+e^{\alpha+\beta x}}, -\infty < x < \infty$$



$$f(x) = \frac{\beta e^{-(\alpha+\beta x)}}{\left[1+e^{-(\alpha+\beta x)}\right]^2} = \frac{\beta e^{\alpha+\beta x}}{1+e^{\alpha+\beta x}}$$



Generalized Linear Models

e. g. 流行病學研究打鼾是否為心臟病的危險因子

		Y	
		1	0
X	$x+1$	$\pi(x+1)$	$1-\pi(x+1)$
	x	$\pi(x)$	$1-\pi(x)$

· $\hat{\beta}$ 的意義：

$\pi(x) = P(Y = 1|X = x)$

$\pi(x+1) = P(Y = 1|X = x+1)$

$$\ln\frac{\pi(x+1)}{1-\pi(x+1)} = \alpha + \beta(x+1)\ ,\ \ln\frac{\pi(x)}{1-\pi(x)} = \alpha + \beta x \Rightarrow \ln\frac{\pi(x+1)}{1-\pi(x+1)} - \ln\frac{\pi(x)}{1-\pi(x)} = \beta$$

$$\therefore \text{odds ratio} = \theta = \frac{\pi(x+1)/[1-\pi(x+1)]}{\pi(x)/[1-\pi(x)]} = e^{\beta} \ \Rightarrow\ \hat{\theta} = e^{\hat{\beta}}$$

即 X 每增加一個單位，$Y = 1$ 對 $Y = 0$ 勝算為原勝算的 $e^{\hat{\beta}}$ 倍。

Generalized Linear Models

e. g. 流行病學研究打鼾是否為心臟病的危險因子

(3) Probit link : $\boldsymbol{\Phi^{-1}[\pi(x)] = \alpha + \beta x \Rightarrow \hat{\pi}(x) = \Phi(\hat{\alpha} + \hat{\beta}x)}$

MLE : $\boldsymbol{\hat{\alpha} = -2.061, \hat{\beta} = 0.188, \text{s.e.}\left(\hat{\beta}\right) = 0.0236, \text{Wald } Z^2 = 63.14}$

· **Let** $\boldsymbol{X \sim N\left(\mu, \sigma^2\right),\ \ \mu = -\frac{\alpha}{\beta} \text{ and } \sigma = \frac{1}{|\beta|}}$

$\boldsymbol{\Rightarrow F(x) = P(X \le x) = P\left(\frac{X-\mu}{\sigma} \le \frac{x-\mu}{\sigma}\right) = \Phi\left(\frac{x-\mu}{\sigma}\right) = \Phi(\alpha + \beta x)}$

· **If** $\boldsymbol{x = -\frac{\alpha}{\beta} \Rightarrow \Phi(\alpha + \beta x) = \Phi(0) = \frac{1}{2}}$

$\boldsymbol{\therefore \text{as } x = -\frac{\hat{\alpha}}{\hat{\beta}} = -\frac{-2.061}{0.188} \approx 11.0\ ,\ \hat{\pi}(x) = \Phi\left(\hat{\alpha} + \hat{\beta}x\right) = \frac{1}{2}}$

· **If** $\boldsymbol{x = 0 \Rightarrow \hat{\pi}(x) = \Phi(\hat{\alpha}) = \Phi(-2.061) \approx 0.0197}$

Generalized Linear Models

***SAS code**

```
options nodate nonotes ps=60;
data glm;
 input snoring disease total;
cards;
0 24 1379
2 35  638
4 21  213
5 30  254
;
proc genmod;
  model disease/total=snoring/dist=bin link=identity;
proc genmod;
  model disease/total=snoring/dist=bin link=logit;
proc genmod;
  model disease/total=snoring/dist=bin link=probit;
run;
```

Generalized Linear Models

***SAS output**

The GENMOD Procedure

Model Information

Description	Value
Data Set	WORK.GLM
Distribution	BINOMIAL
Link Function	IDENTITY
Dependent Variable	DISEASE
Dependent Variable	TOTAL
Observations Used	4
Number Of Events	110
Number Of Trials	2484

Criteria For Assessing Goodness Of Fit

Criteria For Assessing Goodness Of Fit

Criterion	DF	Value	Value/DF
Deviance	2	0.0692	0.0346
Scaled Deviance	2	0.0692	0.0346
Pearson Chi-Square	2	0.0688	0.0344
Scaled Pearson X2	2	0.0688	0.0344
Log Likelihood	.	-417.4960	.

Analysis Of Parameter Estimates

Parameter	DF	Estimate	Std Err	ChiSquare	Pr>Chi
INTERCEPT	1	$\hat{\alpha}$ = 0.0172	0.0034	25.1805	0.0001
SNORING	1	$\hat{\beta}$ = 0.0198	0.0028	49.9708	0.0001 (s)
SCALE	0	1.0000	0.0000	.	.

NOTE: The scale parameter was held fixed.

Generalized Linear Models

*SAS output

The GENMOD Procedure

Description	Value
Data Set	WORK.GLM
Distribution	BINOMIAL
Link Function	LOGIT
Dependent Variable	DISEASE
Dependent Variable	TOTAL
Observations Used	4
Number Of Events	110
Number Of Trials	2484

Criteria For Assessing Goodness Of Fit

Criterion	DF	Value	Value/DF
Deviance	2	2.8089	1.4045
Scaled Deviance	2	2.8089	1.4045
Pearson Chi-Square	2	2.8743	1.4372
Scaled Pearson X2	2	2.8743	1.4372
Log Likelihood	.	-418.8658	.

Analysis Of Parameter Estimates

Parameter	DF	Estimate	Std Err	ChiSquare	Pr>Chi
INTERCEPT	1	$\hat{\alpha}=$ -3.8662	0.1662	541.0562	0.0001
SNORING	1	$\hat{\beta}=$ 0.3973	0.0500	63.1236	0.0001 (s)
SCALE	0	1.0000	0.0000	.	.

NOTE: The scale parameter was held fixed.

Generalized Linear Models

***SAS output**

Description	Value
Data Set	WORK.GLM
Distribution	BINOMIAL
Link Function	PROBIT
Dependent Variable	DISEASE
Dependent Variable	TOTAL
Observations Used	4
Number Of Events	110
Number Of Trials	2484

Criteria For Assessing Goodness Of Fit

Criterion	DF	Value	Value/DF
Deviance	2	1.8716	0.9358
Scaled Deviance	2	1.8716	0.9358
Pearson Chi-Square	2	1.9101	0.9550
Scaled Pearson X2	2	1.9101	0.9550
Log Likelihood	.	-418.3971	.

Analysis Of Parameter Estimates

Parameter	DF	Estimate	Std Err	ChiSquare	Pr>Chi	
INTERCEPT	1	$\hat{\alpha}=$ -2.0606	0.0704	855.4905	0.0001	
SNORING	1	$\hat{\beta}=$ 0.1878	0.0236	63.1431	0.0001	(s)
SCALE	0	1.0000	0.0000	.	.	

NOTE: The scale parameter was held fixed.

Generalized Linear Models

***Poisson Regression (log link)**

$\mathbf{Y \sim Poi(\mu),\ \mu = E(Y)}$

$\mathbf{\ln\mu = \alpha + \beta x \Rightarrow \hat{\mu} = e^{\hat{\alpha}+\hat{\beta}x} = e^{\hat{\alpha}}\left(e^{\hat{\beta}}\right)^{x} = \hat{\pi}(x)}$

· $\hat{\boldsymbol{\beta}}$ 的意義：$\mathbf{\hat{\pi}(x+1) = e^{\hat{\alpha}}\left(e^{\hat{\beta}}\right)^{x+1}}$, $\mathbf{\hat{\pi}(x) = e^{\hat{\alpha}}\left(e^{\hat{\beta}}\right)^{x} \Rightarrow \frac{\hat{\pi}(x+1)}{\hat{\pi}(x)} = e^{\hat{\beta}}}$

(即 X 每增加一個單位，Y 的平均為原本平均的 $e^{\hat{\beta}}$ 倍)

(1) If $\boldsymbol{\beta = 0}$, then $\boldsymbol{\mu = \pi(x) = e^{\alpha}}$: 即 Y 的平均不因 X 改變。

(2) If $\boldsymbol{\beta > 0}$, then $\boldsymbol{e^{\beta} > 1}$: 即 Y 的平均隨 X 增加而增加。

(3) If $\boldsymbol{\beta = 0}$, then $\boldsymbol{e^{\beta} < 1}$: 即 Y 的平均隨 X 增加而減少。

Generalized Linear Models

e. g. 研究影響母螃蟹追求者的因素

Y : 屬於母螃蟹蟹蟹穴附近的公螃蟹數

X : 蟹殼寬度

$\mu = E(Y)$

· Poisson loglinear fit : $\ln\hat{\mu} = \hat{\alpha} + \hat{\beta}x = -3.305 + 0.164x,\ \hat{\beta} = 0.164,\ s.e.(\hat{\beta}) = 0.02$

As $x = 26.3$, $\hat{\mu} = \exp\{\hat{\alpha} + \hat{\beta}\cdot 26.3\} \approx 2.74 = \hat{\pi}(26.3)$

As $x = 27.3$, $\hat{\mu} = \exp\{\hat{\alpha} + \hat{\beta}\cdot 27.3\} \approx 3.23 = \hat{\pi}(27.3) \Rightarrow \frac{\hat{\pi}(27.3)}{\hat{\pi}(26.3)} = e^{\hat{\beta}} = \frac{3.23}{2.74} \approx 1.18$

即蟹殼寬度每增加 1 公分，母蟹的追求者平均增加 18%。

· 去掉 $x = 33.5$ 的資料後：$\hat{\alpha} = -3.461,\ \hat{\beta} = 0.170,\ s.e.(\hat{\beta}) = 0.022$。

Generalized Linear Models

e. g. 研究影響母螃蟹追求者的因素

· **Identity link** : $\hat{\mu} = \hat{\alpha} + \hat{\beta}x = -11.53 + 0.55x,\ \hat{\beta} = 0.55,\ s.e.(\hat{\beta}) = 0.059$

As $x = 26.3$, $\hat{\mu} = -11.53 + 0.55 \cdot 26.3 \approx 2.83 = \hat{\pi}(26.3)$

As $x = 27.3$, $\hat{\mu} = -11.53 + 0.55 \cdot 27.3 \approx 3.48 = \hat{\pi}(27.3)$

即蟹殼寬度每增加 1 公分，母蟹的追求者平均增加 0.55 隻。

Generalized Linear Models

***Poisson Regression for rate data**

Y : 在 t 時間單位或空間單位內事件發生次數

$\frac{Y}{t}$ **: the sample rate of outcomes** $\Rightarrow \frac{\mu}{t} = E(\frac{Y}{t})$

loglinear model : $\ln\frac{\mu}{t} = \alpha + \beta x \Rightarrow \mu = t \cdot e^{\hat{\alpha}+\hat{\beta}x} \Rightarrow \ln\mu - \ln t = \alpha + \beta x$

ln t : offset (補償項)

Generalized Linear Models

e. g. 年長駕車者發生交通事件資料，共 16262 位 65 至 84 歲受訪者，每一位觀察 0 至 4 年，

所有女性被觀察的時間為 17.3 千年，在這期間有 175 件受傷事故；

所有男性被觀察的時間為 21.4 千年，在這期間有 320 件受傷事故。

令 Y : 受傷交通事故件數; $\mathbf{X} = \begin{cases} \mathbf{1}, & \text{男性} \\ \mathbf{0}, & \text{女性} \end{cases}$

$$\boldsymbol{\mu} = \mathbf{E}(\mathbf{Y})$$

男性每千年每筆交通事故平均件數 : $\frac{\mathbf{320}}{\mathbf{21.4}} \approx \mathbf{14.95}$

女性每千年每筆交通事故平均件數 : $\frac{\mathbf{175}}{\mathbf{17.3}} \approx \mathbf{10.12}$

Generalized Linear Models

$$\ln\frac{\mu}{t}=\alpha+\beta x\Rightarrow\begin{cases}\ln 14.95=\hat{\alpha}+\hat{\beta}=2.7\\ \ln 10.12=\hat{\alpha}=2.31\end{cases}\Longrightarrow\hat{\beta}=0.39\quad\therefore e^{\hat{\beta}}=e^{0.39}=1.48=\frac{14.95}{10.12}$$

即男性每千年發生交通事故平均件數較女性多 48% 。

(1) $H_0:\beta=0$ vs. $H_1:\beta\neq 0$

$$\hat{\beta}=0.39,\ \text{s.e.}\left(\hat{\beta}\right)=0.09$$

(2) $Z=\frac{\hat{\beta}}{\text{s.e.}(\hat{\beta})}=\frac{0.39}{0.09}\approx 4.33\Rightarrow Z^2=18.77\sim\chi_1^2$

(3) Since $Z^2=18.77>3.84=\chi^2_{1,0.05}$ $\therefore$ Reject H_0

*當所有事件發生次數都有相同時間單位或空間單位，或發生次數並沒有參考單位時間，model 中可不須 offset 項。

Generalized Linear Models

***Wald test**

$\because \hat{\beta}$: MLE of β $\therefore \hat{\beta} \sim N\left(\beta, \sigma^2_{\hat{\beta}}\right)$, where $\sigma^2_{\hat{\beta}}$ is the asymptotic variance of $\hat{\beta}$

(1) $H_0 : \beta = 0$ vs. $H_1 : \beta \neq 0$

(2) Test statistic $Z = \frac{\hat{\beta}}{\text{s.e.}(\hat{\beta})} \sim N(0,1)$ under $H_0 : \beta = 0 \Rightarrow Z^2 \sim \chi^2_1$

Z **: Wald statistic**

(3) Reject H_0 if $Z^2 > \chi^2_{1,\alpha}$

Generalized Linear Models

*Likelihood-ratio test

(1) $\begin{cases} H_0 : \text{reduce } model \\ H_1 : \text{full model} \end{cases}$ e.g. $\begin{cases} H_0 : \ln\mu = \alpha \\ H_1 : \ln\mu = \alpha + \beta x \end{cases} \Leftrightarrow \begin{cases} H_0 : \beta = 0 \\ H_1 : \beta \neq 0 \end{cases}$

(2) the likelihood-ratio test statistic : $G^2 = -2\ln\left(\frac{\ell_0}{\ell_1}\right) = -2(\ln\ell_0 - \ln\ell_1) = -2(L_0 - L_1) \sim \chi^2_{df}$,

where ℓ_1 denotes the maximized value of the likelihood function for the full model

ℓ_0 denotes the maximized value of the likelihood function

for the simpler model under the null hypothesis

$df =$ the number of parameters in full model $-$

the number of parameters in reduce model

(3) Reject H_0 if $G^2 > \chi^2_{df,\alpha}$

Generalized Linear Models

*Score test

優點：即使無假設 fit model，仍可求 Score statistic。

當 MLE of β 不存在或 $\hat{\beta} = \infty$ 時，仍可求 Score statistic。

(1) $\begin{cases} H_0 : \text{reduce model} \\ H_1 : \text{full model} \end{cases}$ e.g. $\begin{cases} H_0 : \ln \mu = \alpha \\ H_1 : \ln \mu = \alpha + \beta x \end{cases}$

(2) the score statistic $= -2 \ln \left(\frac{\ell_0}{\ell_1}\right) = U'(\vec{\beta}_0) \cdot I^{-1}(\vec{\beta}_0) \cdot U(\vec{\beta}_0) \sim \chi^2_{df}$,

$\vec{\beta} = (\alpha, \beta)$

$\vec{\beta}_0 = (\alpha, 0)$

where $U(\vec{\beta}_0) = \left[\frac{\partial \ln \ell}{\partial \vec{\beta}}\right]_{\vec{\beta}=\vec{\beta}_0}$, $I(\vec{\beta}_0) = \left[\frac{\partial^2 \ln \ell}{\partial \vec{\beta}\, \partial \vec{\beta}'}\right]_{\vec{\beta}=\vec{\beta}_0}$: Fisher's information matrix

df = the number of parameters in full model −

the number of parameters in reduce model

(3) Reject H_0 if score $> \chi^2_{df,0.05}$ or $p - \text{value} < 0.05$

・在實務上，the likelihood-ratio test 較 the Wald test 檢定力高。

Generalized Linear Models

***Criteria for assessing goodness of fit**

(1) $\begin{cases} H_0 : \ln \mu_i = \alpha + \beta x_i, i = 1, 2, \dots, k \\ \quad H_1 : \text{saturated model} \end{cases}$

(2) The Pearson goodness-of-fit or the likelihood-ratio goodness-of-fit :

$$X^2 = \sum_{i=1}^{k} \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i} \sim \chi^2_{df} \text{ or } G^2 = 2\sum_{i=1}^{k} y_i \ln \frac{y_i}{\hat{\mu}_i} \sim \chi^2_{df} \text{ if } \hat{\mu}_i \geq 5 \text{ and } k \text{ is fixed}$$

where k is the number of distinct level of explanatory variables

y_i is the observed count under the i th level of explanatory variables

$\mu_i = E(y_i)$

$df = \text{the number of response counts} - \text{the number of model parameters}$

(3) Reject H_0 if $X^2 > \chi^2_{df,\alpha}$ i.e. poor model fit

Generalized Linear Models

***Criteria for assessing goodness of fit**

e. g. 螃蟹追求者 Poisson Link

· 當 $k = 66$：$X^2 = 174.3$，$G^2 = 190$，$df = 64$

但因 $\hat{\mu}_i < 5$ for many observed total counts 且 k is not fixed，在此 X^2 和 G^2 不可信。

故將資料重新分8類(如後面SAS code)

· 當 $k = 8$：$X^2 = 6.5$，$G^2 = 6.9$，$df = 8 - 2 = 6$， $p - \text{value} > 0.8$

∴ good fit

· 重新分類的方法並不固定，一般分至 6~10 類，但須使每一類下的 $\hat{\mu}_i \geq 5$

Generalized Linear Models

***Criteria for assessing goodness of fit**

e. g. 螃蟹追求者 Poisson Link

· **若令 width scores** $(22.69, 23.84, 24.77, 25.84, 26.79, 27.74, 28.67, 30.41)$

$\Rightarrow \hat{\alpha} = -3.535, \hat{\beta} = 0.173, \text{s.e.}(\hat{\beta}) = 0.021$

As the score of $x = 22.69$**, the estimated expected number of satellites per female crab is**

$\exp\{-3.535 + 0.173 \cdot 22.69\} \approx 1.47$

$\Rightarrow \hat{\mu}_1 = 14 \times 1.47 = 20.5, y_1 = 14$

$X^2 = 6.2, G^2 = 6.5, df = 6, p - \text{value} > 0.3$

$\therefore$ **no evidence of model lack of fit**

Generalized Linear Models

***SAS code**

```
options nodate nonotes ps=60;
data crab;
  input width cases satell;
  log_case=log(cases);
cards;
22.69 14  14
23.84 14  20
24.77 28  67
25.84 39 105
26.79 22  63
27.74 24  93
28.67 18  71
30.41 14  72
;
proc genmod;
  model satell=width/dist=poi link=log
        offset=log_case obstats residuals;
run;
```

Generalized Linear Models

***SAS output**

The GENMOD Procedure

Model Information

Description	Value
Data Set	WORK.CRAB
Distribution	POISSON
Link Function	LOG
Dependent Variable	SATELL
Offset Variable	LOG_CASE
Observations Used	8

Criteria For Assessing Goodness Of Fit

Criterion	DF	Value	Value/DF
Deviance	6	G^2 = 6.5168	1.0861
Scaled Deviance	6	6.5168	1.0861
Pearson Chi-Square	6	X^2 = 6.2470	1.0412
Scaled Pearson X2	6	6.2470	1.0412
Log Likelihood	.	1652.1028	.

Analysis Of Parameter Estimates

Parameter	DF	Estimate	Std Err	ChiSquare	Pr>Chi
INTERCEPT	1	-3.5355	0.5760	37.6698	0.0001
WIDTH	1	0.1727	0.0212	66.1750	0.0001
SCALE	0	1.0000	0.0000	.	.

NOTE: The scale parameter was held fixed.

Generalized Linear Models

***SAS output**

Observation Statistics

SATELL	Pred	Xbeta	Std	HessWgt	Lower	Upper
14	20.5444	0.3835	0.1027	20.5444	16.7986	25.1253
20	25.0585	0.5822	0.0814	25.0585	21.3639	29.3921
67	58.8498	0.7428	0.0657	58.8498	51.7349	66.9433
105	98.6084	0.9276	0.0514	98.6084	89.1625	109.0549
63	65.5439	1.0917	0.0448	65.5439	60.0296	71.5647
93	84.2522	1.2558	0.0469	84.2522	76.8599	92.3555
71	74.1998	1.4164	0.0563	74.1998	66.4541	82.8484
72	77.9431	1.7169	0.0841	77.9431	66.0995	91.9088

Observation Statistics

Resraw	Reschi	Resdev	StResdev	StReschi	Reslik
-6.5444	-1.4438	-1.5329	-1.7320	-1.6314	-1.7107
-5.0585	-1.0105	-1.0477	-1.1473	-1.1065	-1.1406
8.1502	1.0624	1.0392	1.2035	1.2304	1.2104
6.3916	0.6437	0.6369	0.7405	0.7484	0.7426
-2.5439	-0.3142	-0.3163	-0.3394	-0.3372	-0.3391
8.7478	0.9530	0.9372	1.0381	1.0556	1.0414
-3.1998	-0.3715	-0.3742	-0.4278	-0.4246	-0.4270
-5.9431	-0.6732	-0.6820	-1.0180	-1.0048	-1.0107

Pearson Residual

Adjusted Residual（負的顯示低估）

Generalized Linear Models

- Pearson residual $= \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}} = \frac{(\text{observed} - \text{fitted})}{\sqrt{\widehat{\text{var}}(\text{observed})}} = e_i$

  the Pearson goodness-of-fit statistic : $X^2 = \sum_{i=1}^{k} \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i} = \sum_{i=1}^{k} e_i^2$

- Adjusted residual $= \frac{e_i}{\text{s.e.}(e_i)} \approx N(0, 1)$ as μ_i is large.

- 當 $y_i - \hat{\mu}_i$ 較大時，利用 adjusted residuals 較易顯現出來。

- 若 $\left|\frac{e_i}{\text{s.e.}(e_i)}\right| > 2$，則第 i 個觀測值 fit 不好，值得注意。

- 一般 $y_i - \hat{\mu}_i < y_i - \mu_i$，Adjusted residual 較常被使用。

Generalized Linear Models

- For Poisson GLMs, the general form of the adjusted residual is $\frac{e_i}{s.e.(e_i)} = \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i(1-h_i)}} = \frac{e_i}{\sqrt{1-h_i}}$,

  where h_i is called the leverage (槓桿點) of observation i .

- $\ln \mu_i = \alpha + \beta x_i$, $h_i = \frac{1}{k} + \frac{(x_i - \bar{x})^2}{\sum_{j=1}^{k}(x_j - \bar{x})^2}$, $i = 1, 2, \ldots, k.$

- h_i 越大代表第 i 個觀測值對模型配適 (the fit of the model) 的影響越大。